

The leading level error of the two-sample Welch statistic is governed
by the smaller group

Note — derivation from the expansion of Zhou and Dinh (2005), not part of the vignette

Starting point

Let $Y_{11}, \dots, Y_{1n_1}$ and $Y_{21}, \dots, Y_{2n_2}$ be independent samples from distributions with means M_1, M_2 , variances V_1, V_2 and skewnesses γ_1, γ_2 , let $N = n_1 + n_2$ and let

$$\lambda = \frac{n_1}{N} \quad (1)$$

be the share of the total sample held by the first group. Let T denote the unpooled two-sample t -statistic, that is Welch's statistic for $k = 2$.

Proposition 1 of Zhou and Dinh [-@Zhou:2005], proved there under the regularity conditions of Hall (1992a) and assuming $\lambda_N = \lambda + O(N^{-r})$ for some $r \geq 0$, states

$$P(T \leq x) = \Phi(x) + \frac{1}{\sqrt{N}} \frac{A}{6} (2x^2 + 1) \varphi(x) + O(N^{-\min(1, r+1/2)}), \quad (2)$$

where Φ and φ are the standard normal distribution and density functions and

$$A = \left(\frac{V_1}{\lambda} + \frac{V_2}{1-\lambda} \right)^{-3/2} \left(\frac{V_1^{3/2} \gamma_1}{\lambda^2} - \frac{V_2^{3/2} \gamma_2}{(1-\lambda)^2} \right). \quad (3)$$

The departure of the distribution of T from normality is thus of order $N^{-1/2}$ with the coefficient $A/6$, and Eq. (3) shows that this coefficient carries both the skewnesses and the imbalance: it vanishes when the two groups agree in size, variance and skewness, since then the two terms of the second factor are equal.

The limit of a small first group

Everything below is a consequence of Eq. (3); it is not stated by Zhou and Dinh [-@Zhou:2005].

Consider $\lambda \rightarrow 0$, the first group becoming a vanishing share of the total. Factor the first bracket of Eq. (3),

$$\frac{V_1}{\lambda} + \frac{V_2}{1-\lambda} = \frac{V_1}{\lambda} \left(1 + \frac{V_2}{V_1} \cdot \frac{\lambda}{1-\lambda} \right), \quad (4)$$

so that, by the binomial series for the exponent $-3/2$,

$$\left(\frac{V_1}{\lambda} + \frac{V_2}{1-\lambda} \right)^{-3/2} = \left(\frac{\lambda}{V_1} \right)^{3/2} \left(1 - \frac{3}{2} \cdot \frac{V_2}{V_1} \lambda + O(\lambda^2) \right). \quad (5)$$

In the second bracket the term in λ^{-2} dominates the term in $(1-\lambda)^{-2}$,

$$\frac{V_1^{3/2}\gamma_1}{\lambda^2} - \frac{V_2^{3/2}\gamma_2}{(1-\lambda)^2} = \frac{V_1^{3/2}\gamma_1}{\lambda^2} \left(1 - \frac{V_2^{3/2}\gamma_2}{V_1^{3/2}\gamma_1} \cdot \frac{\lambda^2}{(1-\lambda)^2} \right) = \frac{V_1^{3/2}\gamma_1}{\lambda^2} (1 + O(\lambda^2)), \quad (6)$$

provided $\gamma_1 \neq 0$. Multiplying Eq. (5) by Eq. (6), the factor $V_1^{3/2}$ cancels and $\lambda^{3/2}\lambda^{-2} = \lambda^{-1/2}$, so that

$$A = \frac{\gamma_1}{\sqrt{\lambda}} \left(1 - \frac{3}{2} \cdot \frac{V_2}{V_1} \lambda + O(\lambda^2) \right). \quad (7)$$

Dividing by $\sqrt{N}$ as it appears in Eq. (2), and using $n_1 = \lambda N$ from Eq. (1),

$$\frac{A}{\sqrt{N}} = \frac{\gamma_1}{\sqrt{\lambda N}} \left(1 - \frac{3}{2} \cdot \frac{V_2}{V_1} \lambda + O(\lambda^2) \right) = \frac{\gamma_1}{\sqrt{n_1}} \left(1 - \frac{3}{2} \cdot \frac{V_2}{V_1} \lambda + O(\lambda^2) \right). \quad (8)$$

Reading

Eq. (8) says that as one group becomes small relative to the other, the leading term of Eq. (2) is governed by that group alone: its skewness divided by the square root of its own size, with the total sample size and the size of the other group entering only through the correction of order λ . Enlarging the second group does not reduce the leading error.

The correction term is negative for $V_2 < \infty$, so the limit is approached from below, and its size is set by the variance ratio: for $V_1 = V_2$ the first correction is $-\frac{3}{2}\lambda$.

Numerical check

For $V_1 = V_2 = 1$, $\gamma_1 = \gamma_2 = 2$ and $N = 1000$, the ratio of $A/\sqrt{N}$ from Eq. (3) to $\gamma_1/\sqrt{n_1}$ is

λ	0.10	0.05	0.02	0.01
ratio	0.843	0.923	0.970	0.985
$1 - \frac{3}{2}\lambda$	0.850	0.925	0.970	0.985

confirming both the limit and the coefficient of the first correction in Eq. (8).

Scope

Eq. (2) is a two-sample result. Nothing here extends it to Welch's F statistic for $k > 2$ groups, for which no corresponding expansion was found in the literature; the requirement reported by Delacre et al. [-@Delacre:2019] that groups hold at least 50 observations for up to four groups, and 100 beyond that, rests on simulation.

References